Final Year Project (Year 22)

Feasibility Study

Electronic and Telecommunication Engineering

A MULTI-RF HEAD RECEIVER ARRAY FOR SENSING AND COMMUNICATIONS

Supervisors :

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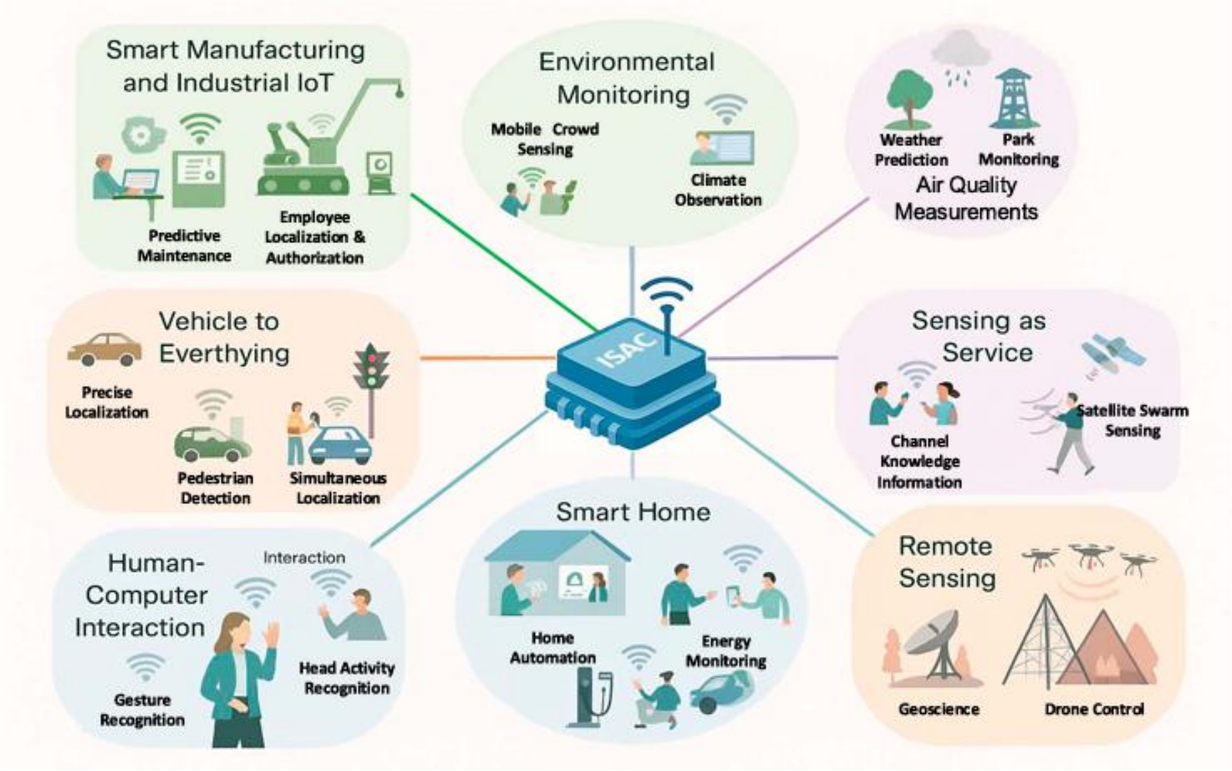
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Motivation

Future promising **6G** communication model expects seamless **connectivity** and **awareness of its surroundings**.



Integrated sensing and communication (ISAC) can enable simultaneous communication and sensing functionality, promising seamless connectivity[1].

[1] K. Wu, Y. Bigdeli, S. A. Keivaan, J. Deng and P. Burasa, "Integrated Sensing and Communication (ISAC) Transceiver: Hardware Architectures, Enabling Technologies, and Emerging Trends," in IEEE Journal of Selected Topics in Electromagnetics, Antennas and Propagation, vol. 1, no. 1, pp. 37-64, Sept. 2025.

Motivation - What is ISAC ?

ISAC - Integrated Sensing and Communication,

- employs a **shared hardware platform**
- utilize **shared waveform**
- **unified signal processing** chain

to perform simultaneous sensing and wireless data communication[1].

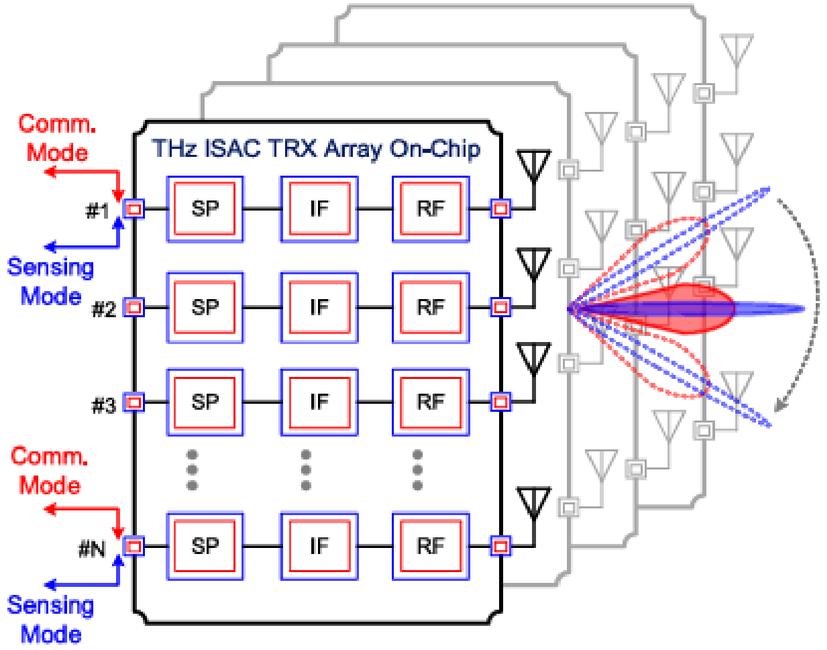


Illustration of ISAC concept

[1] K. Wu, Y. Bigdeli, S. A. Keivaan, J. Deng and P. Burasa, "Integrated Sensing and Communication (ISAC) Transceiver: Hardware Architectures, Enabling Technologies, and Emerging Trends," in IEEE Journal of Selected Topics in Electromagnetics, Antennas and Propagation, vol. 1, no. 1, pp. 37-64, Sept. 2025.

Problem Statement

Current ISAC receivers lack flexible multi-band communication capability while efficiently sharing common hardware [1].

Why do we need multiple bands ?

- It’s difficult to optimize hardware for both sensing and communication over a wide continuous bandwidth [1].
- But it’s possible to optimize for fragmented bands [1].

[1] K. Wu, Y. Bigdeli, S. A. Keivaan, J. Deng and P. Burasa, "Integrated Sensing and Communication (ISAC) Transceiver: Hardware Architectures, Enabling Technologies, and Emerging Trends," in IEEE Journal of Selected Topics in Electromagnetics, Antennas and Propagation, vol. 1, no. 1, pp. 37-64, Sept. 2025.

Primary Objectives

Building a **multi-head radio frequency (RF) chain** and **three antenna arrays** for both sensing and communication without much hardware redundancy.

Implement **simultaneous communication and unmanned aerial vehicle(UAV) detection**.

Scope

Multi-head RF receiver

- Simulate and design of **antenna arrays (two Vivaldi arrays and one patch antenna array)**.
- Simulate and design of RF chains for communication and sensing.
- Perform testing and verify the functionality of the receiver in the field.

Communication and sensing implementation

- Verify sensing capabilities using orthogonal time frequency space modulation (**OTFS**).
- Enable **multi-band communication**.

Out of Scope

Analog or digital beam-forming

- **Beam-forming** is excluded because it requires **multiple synchronized RF chains** and significantly i**ncreases cost, hardware complexity, calibration**, and processing requirements; it is considered as a future extension.

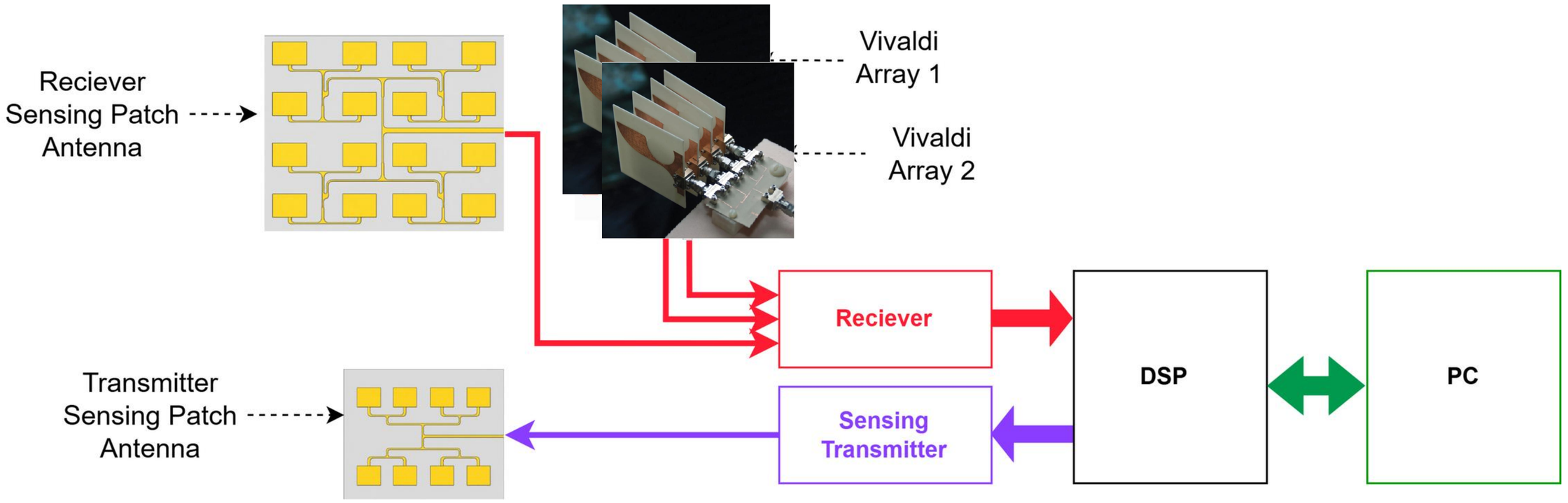
Project deliverables

- **Multi-RF head receiver array** with a single RF channel PCB, antennas, and RF front ends.
- **Embedded software and processing** for receiver control, configuration, and received-data processing.
- **Conference or journal paper** submitted to a recognized publication venue.

Literature review

Paper	
Pegoraro et al., “HiSAC: High-Resolution Sensing with Multiband Communication Signals”	can
Wang et al., “An RFSoC-Based Scalable OTFS Prototyping Platform for Integrated Sensing and Communications”	Real OTF

Project architecture



Current Project Architecture

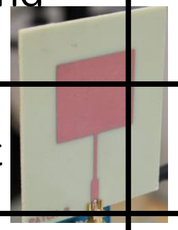
Antenna arrays

Antenna array	Operating frequency range	Bandwidth	Center freq
Vivaldi array 1	800–1600 MHz	800 MHz	1.2 GHz

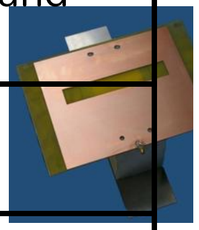
Patch antenna design

Why do need to use a patch antenna?

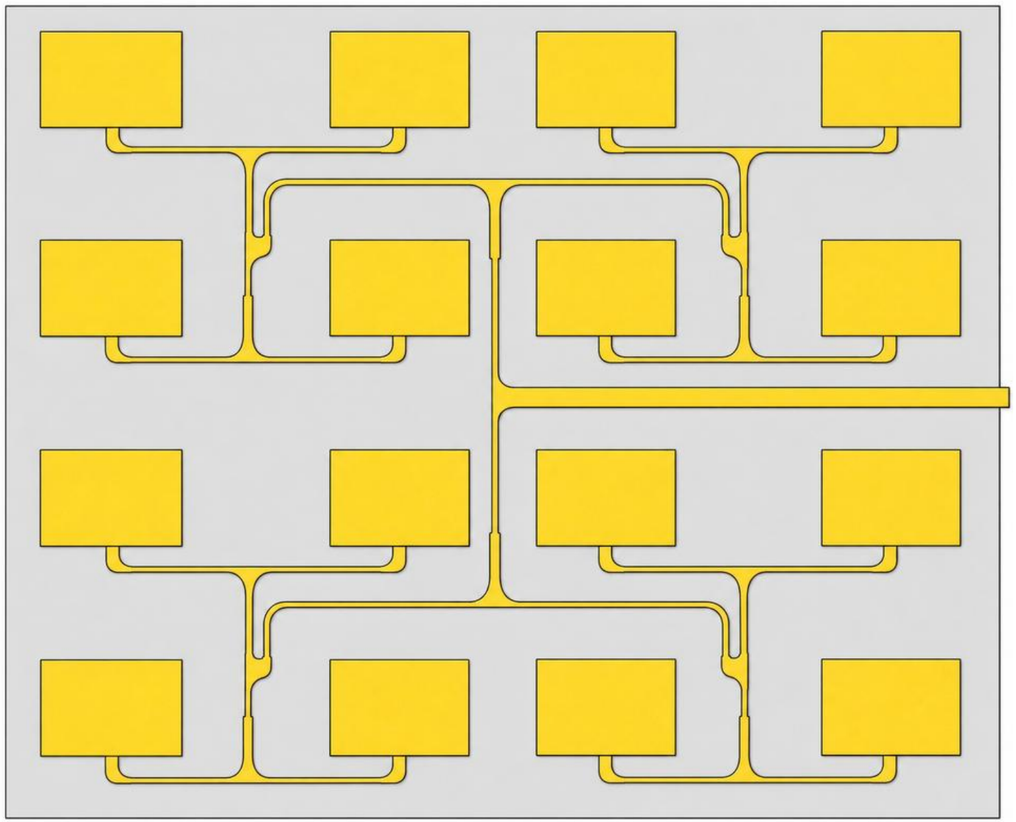
Basic structure	ground plane	directors	ground plane
Bandwidth	Naturally narrowband	Narrow to moderate bandwidth	Narrowband or broadband
Physical profile	Low and compact	Longer and protruding	Low and planar
PCB integration/ Fabrication	Excellent/ simple	Normally requires separate mechanical structure	Excellent/ PCB-compatible
Main advantage	Compact, planar, and easy to integrate	High directional gian	Low profile with flexible geometry







Patch antenna design



Proposed patch antenna array design

Why do we use an antenna array of 16 element patches ?

- The proposed design is a ***16-element (4 × 4) corporate edge-fed*** microstrip patch antenna array, impedance-matched using ***quarter-wave transformers***.
- An antenna array provides a **higher directivity** and **gain, narrower beamwidth**, **controllerable beam direction** and **control over sidelobes and nulls** etc.
- Ideally, ***D_"array" ≈ N * D_"element"*** **.**
- A 4 x 4 array provides benefits such as ***two dimensional beam control***, ***two-axis steering***, ***balanced E/H beamwidths***, ***compact aspect ratio*** etc.

Patch antenna design

Evolution of patch antenna array design and simulation ($f_0 = 5.85GHz$ $\epsilon_r = 4.4$)

1. *Patch-width equation (W)*

$$W = \frac{c}{2f_o\sqrt{\frac{\varepsilon_R+1}{2}}}$$

2. *Effective Permittivity (ε_{eff})*

$$\varepsilon_{eff} = \frac{\varepsilon_R+1}{2} + \frac{\varepsilon_R-1}{2}\left[\frac{1}{\sqrt{1+12\left(\frac{h}{W}\right)}}\right]$$

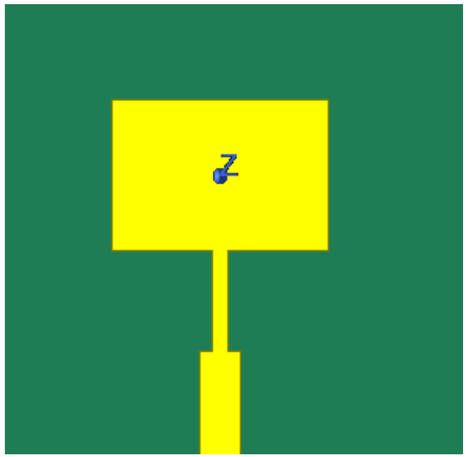
3. *Patch-length equation (L)*

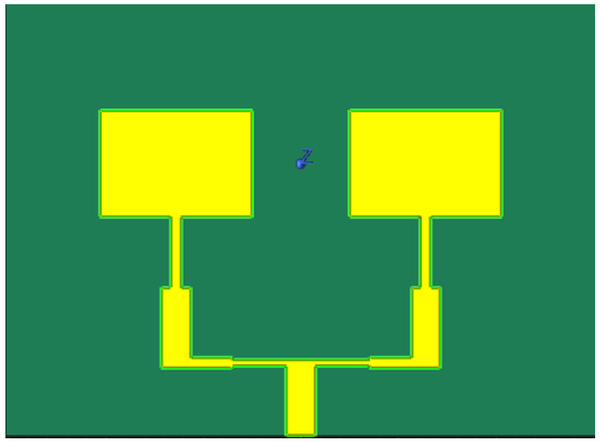
$$L = \frac{c}{2f_o\sqrt{\varepsilon_{eff}}} - 0.824h\left(\frac{(\varepsilon_{eff}+0.3)\left(\frac{W}{h}+0.264\right)}{(\varepsilon_{eff}-0.258)\left(\frac{W}{h}+0.8\right)}\right)$$

4. *Input-impedance equation (Z_{in})*

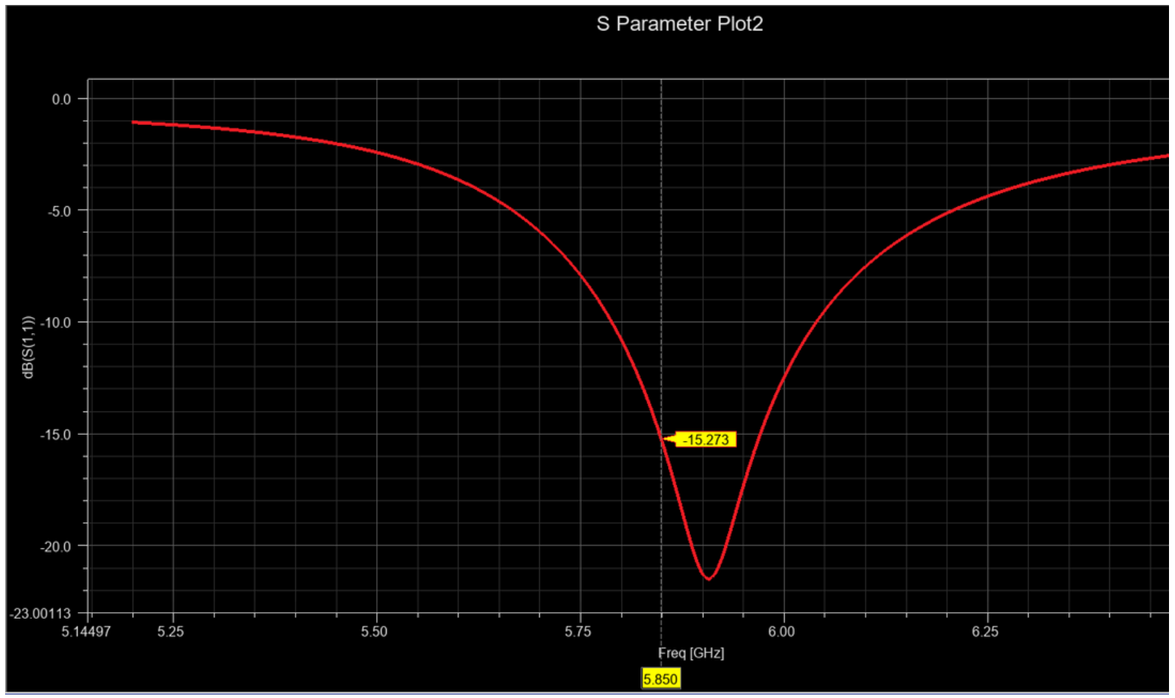
$$Z_{in} = 90\left(\frac{\varepsilon_r^2}{\varepsilon_r - 1}\right)\left(\frac{L}{W}\right)^2$$

Design equations and parameter values



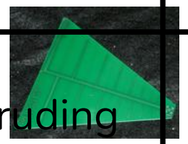


Tuned single and dual patch elements designs

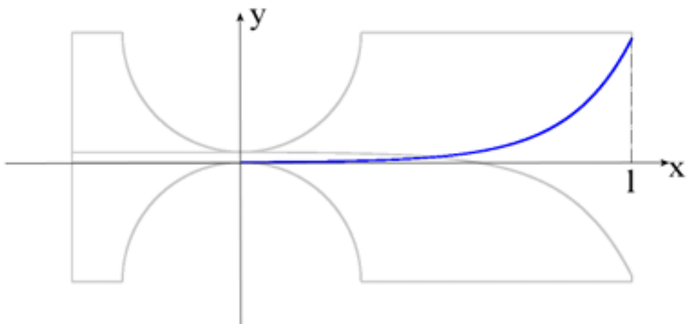


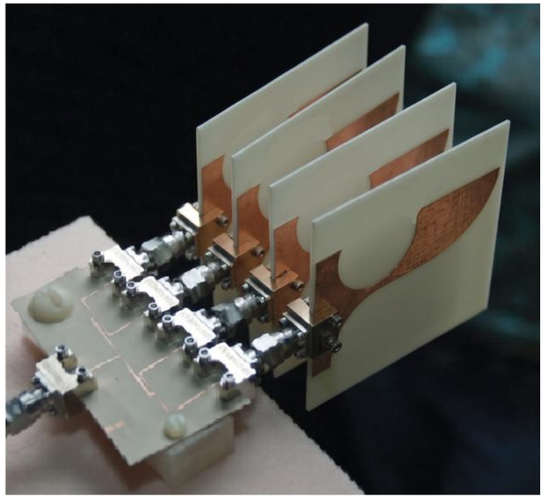
Antipodal Vivaldi antenna design

Why do need to use a Vivaldi antenna?

Basic structure	waveguide	and parasitic elements	tapered slot	conductors with tapered slot
Bandwidth	Very wide	Very wide	Ultra-wideband	Ultra-wideband
Physical profile	Bulky and large 	Long and protruding 	Low-profile and planar 	Low-profile and planar 
PCB integration/ fabrication	Difficult	Moderate	PCB-compatible	PCB-compatible
Main advantage	High gain and directivity	Very wide frequency coverage	Wideband and low profile	Wideband, compact

Antipodal Vivaldi antenna design





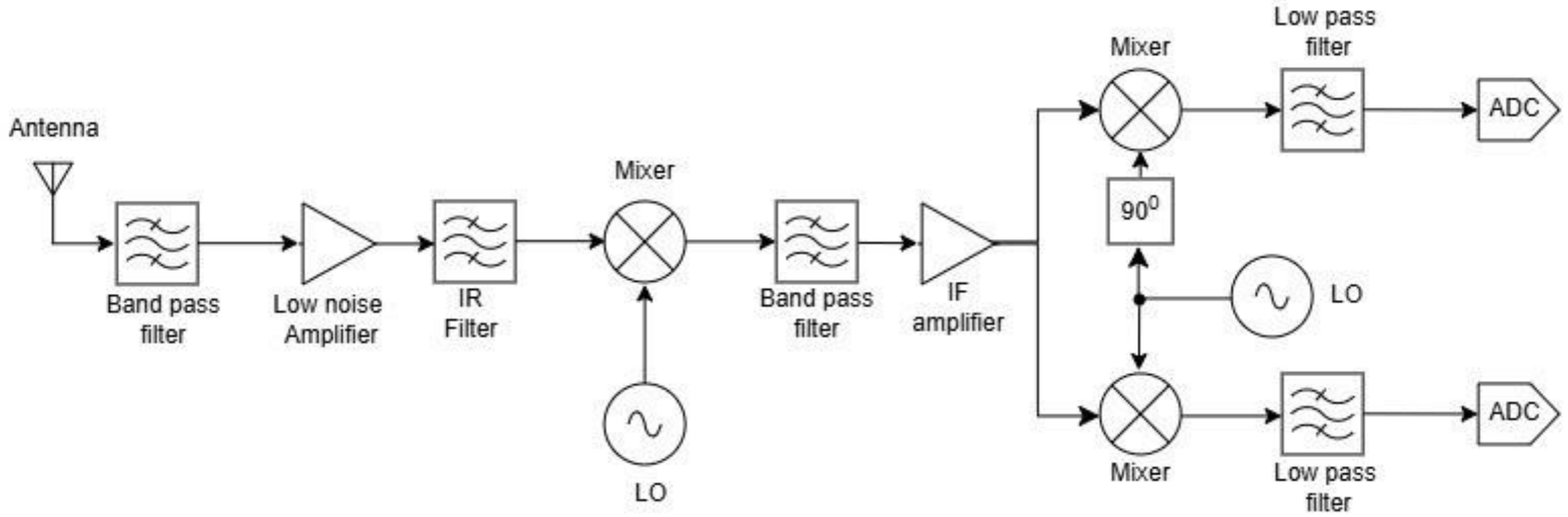
Why do we use an antipodal Vivaldi antenna?

- Wide-band operation with broad impedance and gain bandwidth.
- Having higher directivity due to tappered slot structure provide end-fire radiation.
- Smooth feed-to-radiator transition for improved impedance matching.

Antenna array: Higher resulting gain of main lobe and can obtain lower side lobes.

G. Vinci and R. Weigel, "Multiband Planar Vivaldi Antenna for Mobile Communication and Industrial Applications," 2010 International Conference on Electromagnetics in Advanced Applications, Sydney, NSW, Australia, 2010, pp. 93-96

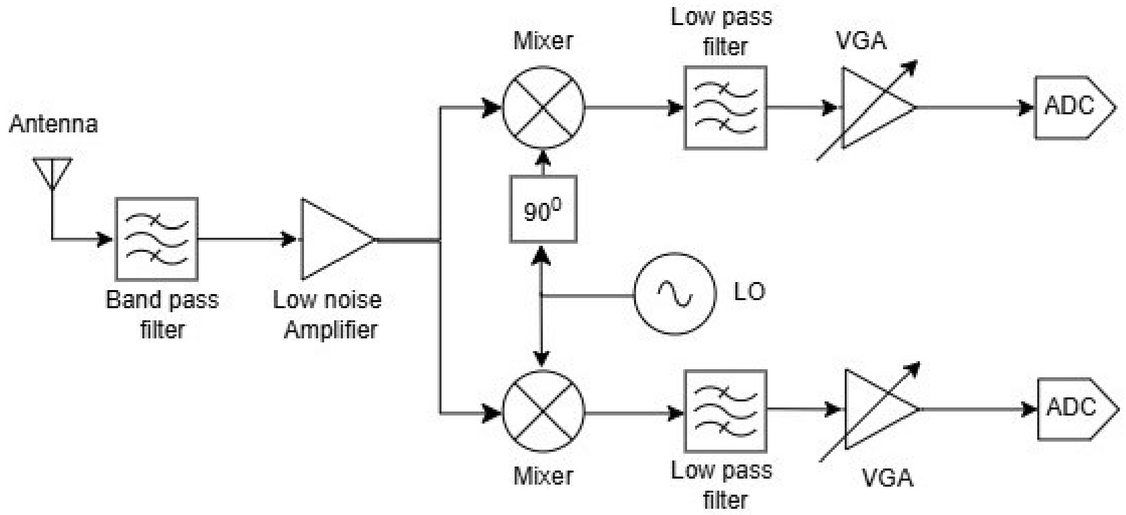
Hetrodyne architecture



- High selectivity through RF/IF filtering and amplification.
- Multiple frequency conversions provide flexible gain distribution and good interference rejection.
- Need to apply additional filters for image-rejection.

B. Razavi, “RF Microelectronics,” 2nd Edition, Prentice Hall Communications Engineering and Emerging Technologies Series, 2012.

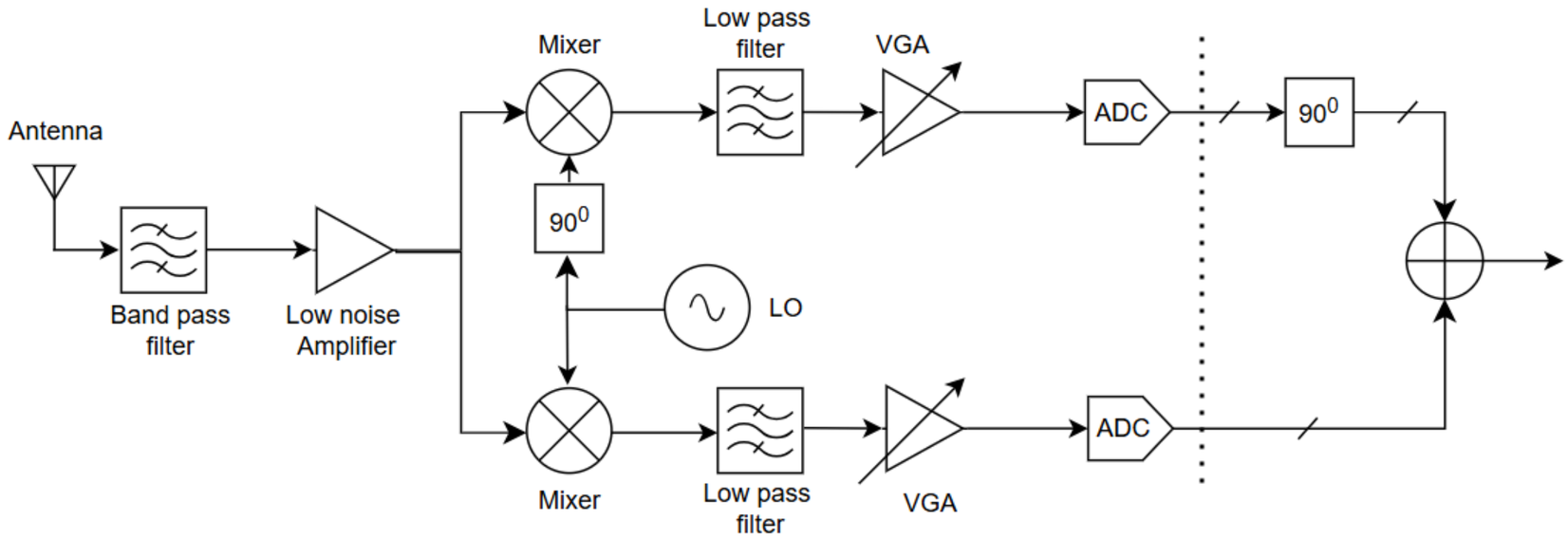
Direct conversion architecture



- Directly converts RF to baseband.
- Simple, low-power, and highly integrated architecture suitable for multi-standard applications.
- DC offset, flicker noise and IQ mismatch can degrade the received signal.

B. Razavi, “RF Microelectronics,” 2nd Edition, Prentice Hall Communications Engineering and Emerging Technologies Series, 2012.

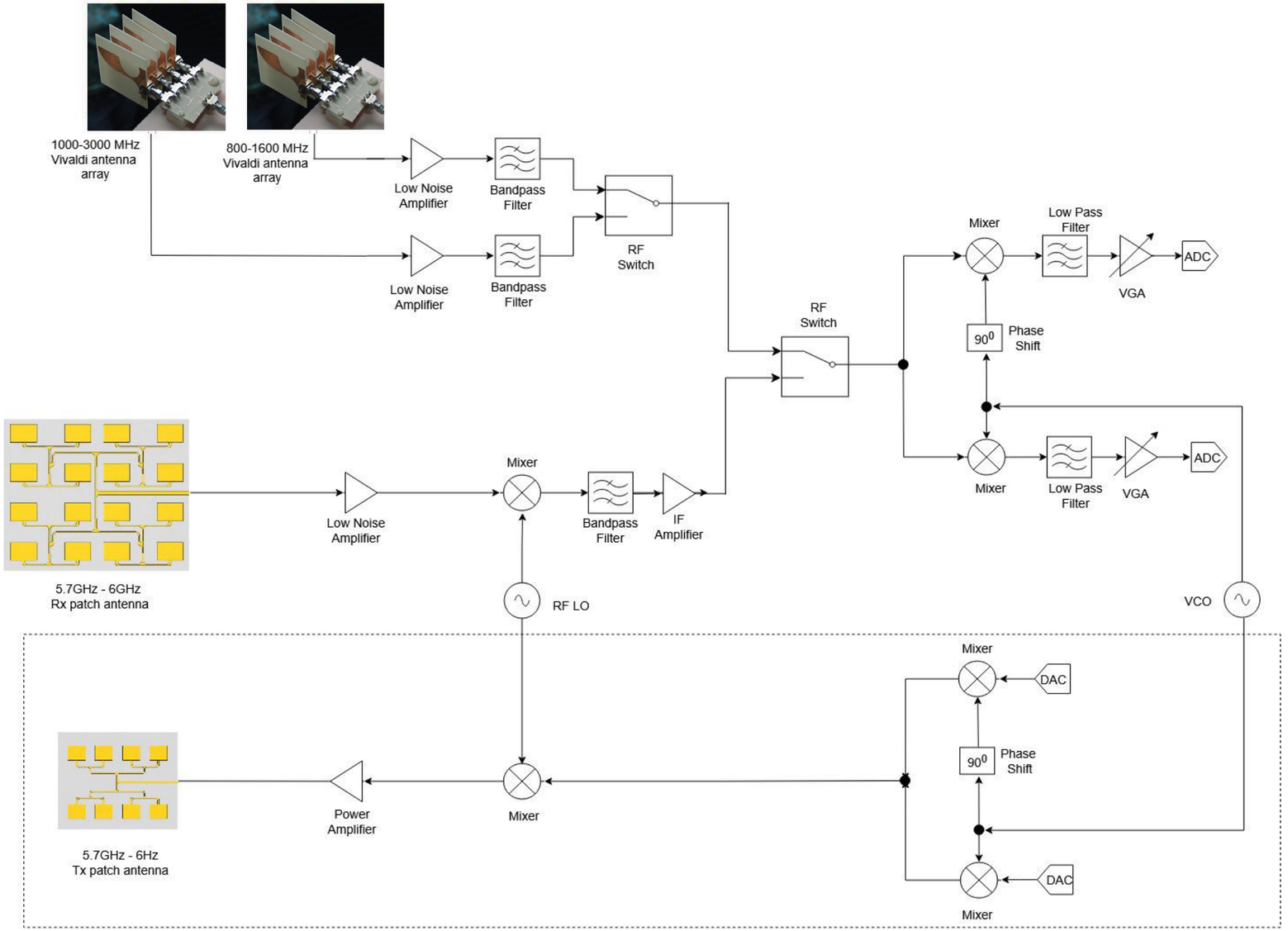
Low IF architecture



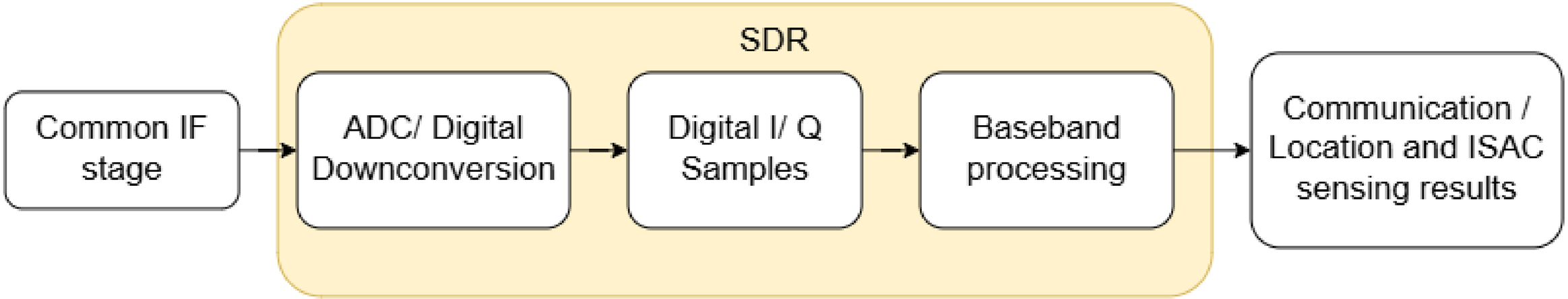
- Converts RF to a low intermediate frequency.
- Image frequencies reappear because the IF is non-zero.
- Image-rejection filtering or mixers are required, increasing implementation complexity.

B. Razavi, “RF Microelectronics,” 2nd Edition, Prentice Hall Communications Engineering and Emerging Technologies Series, 2012.

Receiver Architecture (RF chain)



Digital signal processing



Three baseband processing paths

- 800 - 1600MHz : Processes GSM/GMSK signals for location intelligence.
- 1000 - 3000MHz : Processes cellular signals.
- OTFS processor : Processes ISFFT- OTFS signals in the 5.7GHz - 6.0GHz band.

What is OTFS ?

The problem with standard signals,

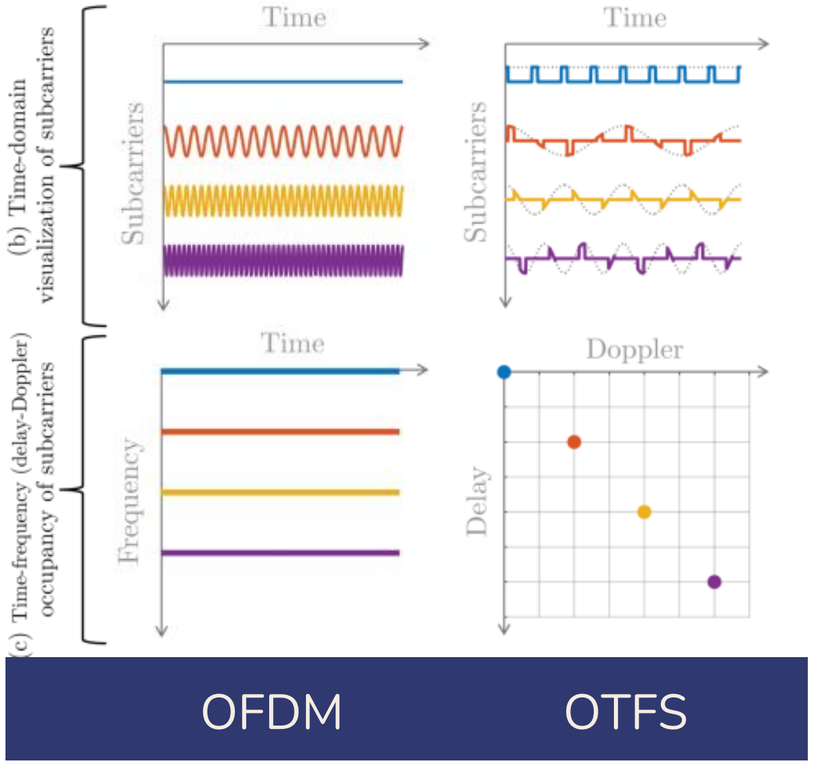
State-of-the-art Orthogonal Frequency Division Multiplexing (OFDM) is **unstable to severe Doppler shifts** which cause Inter-Carrier Interference (ICI).

- e.g., fast moving targets like drones creates sever Doppler shifts

OTFS solves this by mapping the data into the Delay-Doppler domain.

How it works?

It modulates information onto a localized pulse in Delay-Doppler domain, which immunes for larger Doppler shifts.



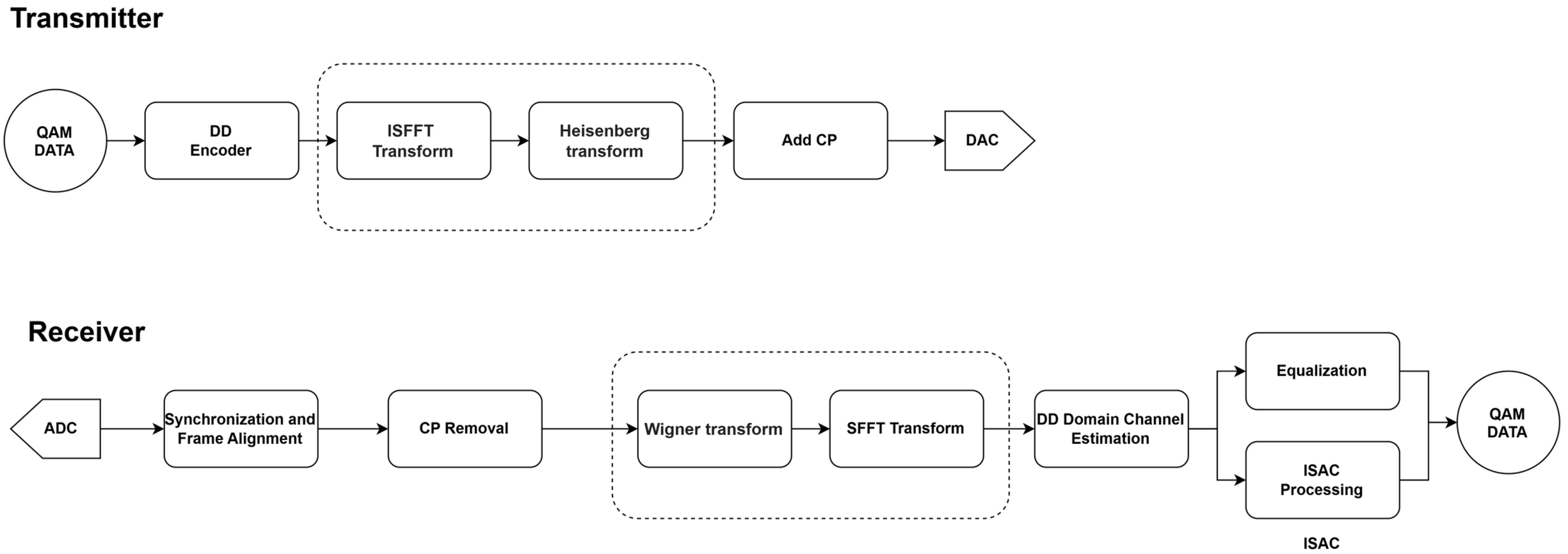
represntation of OFDM and OTFS on time and time frequency domain

Therefore OTFS become prominent in future highly dynamic communication topologies.

Delay-Doppler domain - A coordinate system that maps channel paths depending on their path delay and Doppler shifts

Rou, Hyeon Seok & Ranasinghe, Kuranage Roche Rayan & Savaux, Vincent & Abreu, Giuseppe & G., David & Masouros, Christos. (2026). Affine Frequency Division Multiplexing (AFDM) for 6G: Properties, Features, and Challenges. IEEE Communications Standards Magazine. 10. 216-225. 10.1109/MCOMSTD.2025.3643183.

ISFFT-OTFS baseband processing block diagram



Xiang, J., Hao, S. & Qi, L. A BEM-OTFS channel estimation method for high mobility 6G-V2X. SIViP 18, 6799–6806 (2024). https://doi.org/10.1007/s11760-024-03352-5
https://in.mathworks.com/help/comm/ug/otfs-modulation.html

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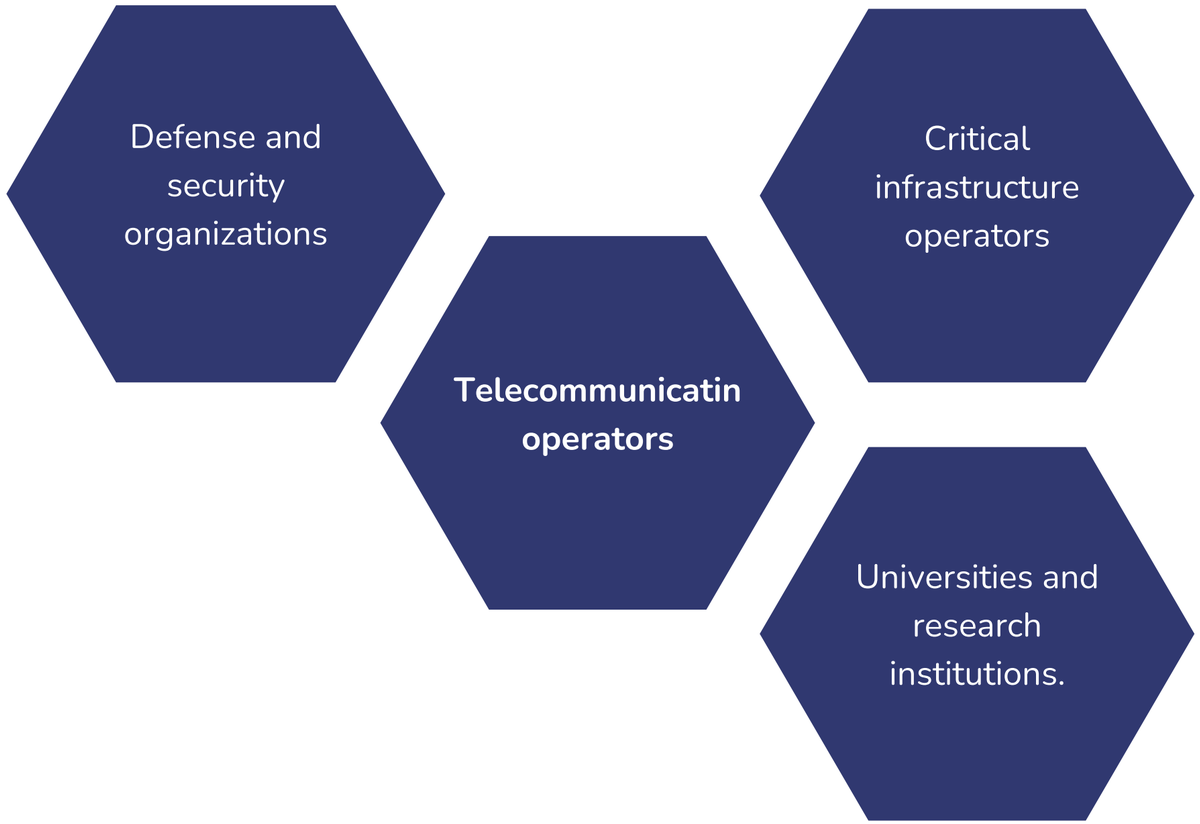
Multifunctional receiver platform

1. Can accommodate multiple communication bands

- e.g.,
  - 800MHz - 1600MHz band - GSM communication
  - 1000MHz - 3000MHz band - Wi-Fi and Bluetooth communication

1. OTFS modulation is used for sensing application

Beneficiaries



Risk factors

- Mutual coupling & RF interference
  - Risk: nearby antennas interfere with each other.
  - Mitigation: optimize spacing and isolation.
- Time constraints
  - Risk: delay in component procurement, manufacturing, or model convergence.
  - Mitigation: parallel task execution and phased integration to reduce bottlenecks.
- Legal / regulatory feasibility
  - Risk: transmitter power may surpass the power limitations imposed by the TRCSL.
  - Mitigation: carefully design the system not to violate power regulations.
- Hardware failure
  - Risk: PCB fabrication defects, antenna mismatches, or component damage.
  - Mitigation: simulate all designs thoroughly and fabricate from certified company .

Bill of Materials

Component	Part	Qty
Mixer	**HMC219BMS8GE**	3
Tx I/Q Modulator	**ADL5375-05ACPZ-R7**	1
Low Noise Amplifier	**HMC717ALP3E**	1
Receiver Mixer	**HMC219BMS8GE**	1
IF Amplifier	**ADL5611ARKZ-R7**	1
Vivaldi-head LNA	**ADL5523ACPZ-R7**	2
RF Head Selector	**SKY13373-460LF**	1
Common I/Q Receiver	**LTC5594IUH#PBF**	1
Fixed 4 GHz LO	**CVCO55CC-4000-4000**	1
Tunable 800–3000 MHz LO	**ADF4351BCPZ**	1
ADCs		2
DACs		2
Passive Components		
PCB , Antenna Array		
Total		

Task delegation

Tasks	Agrajith	Bathiya	Dulina	Ishan
Antena design and simulation				
ISAC algorithm design				
SDR implementation				
Receiver design				
PCB Design				
General				
Documentation				

Resource Requirements

Software tools

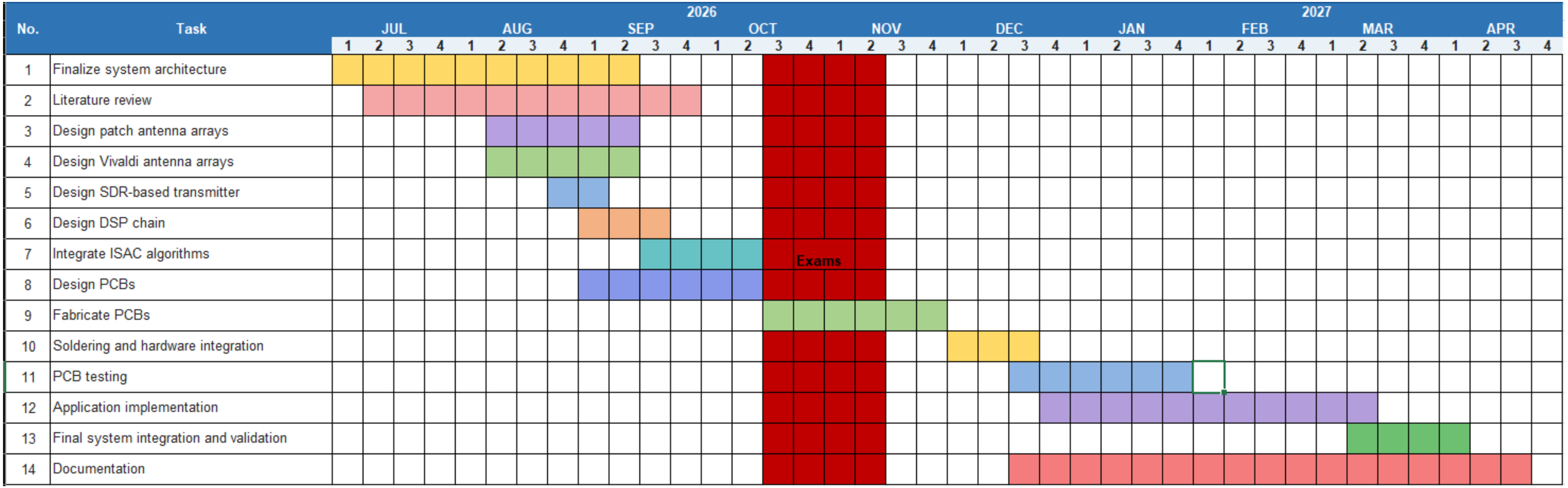
- RF simulation tools (ADS, HFSS), Altium PCB designer.
- Xilinx Vivado/ Vitis HLS.
- MATLAB/Python-based signal processing.

Hardware tools

- RF LNAs, mixers, switches, tunable filters, and IF amplifiers
- Spectrum analyzer, VNA, signal generator, and RF measurement equipment
- FPGA , SDR for digital signal processing

LNA- Low Noise Amplifier , IF - Intermediate Frequency , VNA - Vector Network Analyzer , SDR - Software Define Radio , FPGA - Field Programmable Gate Array

Project timeline



MORE INFO

Lack of multiband flexibility

Ref.	Paper	Cat.	Fabrication Process	Freq. (GHz)	Data Rate (Gb/s)	Mod. Mode	Com. BW (GHz)	TX Pout (dBm)	Ang. Res. (°)	Ran. Res. (m)	Max Ran. (m)	RX NF (dB)
[19] 2012	24 GHz Time Agile ISAC	MB*	PCB	24	0.050	BPSK/ QPSK	—	—	—	1.65	70	—
[99] 2016	60 GHz 6-Ch AiP TRX	MB	0.13 μ m SiGe	57–64	0.4	OOK	7	≈ 4	—	0.02	10	< 10
[24] 2019	40 GHz Sourceless RX	MB	65 nm CMOS	40	0.004	QPSK/16-QAM	7	—	—	—	—	—
[100] 2021	Reconfigurable 35 GHz RX	MB	90 nm CMOS	35	—	—	3.4	16.9	—	—	200	7.6
[88] 2021	THz Comb Radiator & RX Imaging	MB	90 nm SiGe BiCMOS	1140	10	64-QAM	—	- 11	—	—	0.35	44
[101] 2022	IQ TRX for D-band JRC	MB	130 nm BiCMOS	110–170	—	BPSK	—	13	—	—	—	—
[25] 2023	D-Band JRC CMOS TRX	MB	28 nm CMOS	122–168	10	16-QAM	20	13	—	0.0125	25	10
[102] 2023	mmW Freq.-Mod. Tx Array for JCS	MB	45 nm CMOS	28–31	3	64-QAM	0.5	18	4	—	—	24.5–36
[103] 2023	60 GHz Phase-Time Array TX	MB	45 nm CMOS	25.5–32.5	36	64-QAM	—	17.5	< 5	—	—	—
[29] 2024	Joint MW/THz Interferometric RX	MPI*	PCB	5.8 & 150	—	M-QAM	N/R	—	—	—	—	—
[27] 2024	Ka-Band 4T/4R TRX	MB	65 nm CMOS	26–34	0.8	64-QAM	5	> 10	—	—	—	≈ 5
[104] 2024	Over-the-Air 26 GHz RX	MB	22 nm FDSOI	26	—	—	0.8	—	—	—	—	—
[61] 2025	6–11 GHz Digital-Intensive SoC	MB	28 nm CMOS	6–11	0.027	BPSK & ToF	2.4	14	4	0.1	10	17
[32] 2025	60 GHz Dual-Mode Gilbert TRX	MB	65 nm CMOS	55–65	7	16-QAM	—	16	—	0.0375	4	5.8
[28] 2025	Waveguide Receiver Array	MPI	Waveguide + 65 nm CMOS	26–30	—	M-QAM	4	—	—	—	—	—
[33] 2025	VTM AoA & Polarization TRX	MPI	PCB	25.4	0.25	64/128-QAM	—	—	0.9	—	—	—
[31] 2025	Joint 4-D Radar & Comm.	MPI	PCB	24	—	64-QAM	—	—	< 1	< 1.5	—	—

Note: MB* = Mixer Based, MPI* = Multiport Interferometric.

[1] K. Wu, Y. Bigdeli, S. A. Keivaan, J. Deng and P. Burasa, "Integrated Sensing and Communication (ISAC) Transceiver: Hardware Architectures, Enabling Technologies, and Emerging Trends," in IEEE Journal of Selected Topics in Electromagnetics, Antennas and Propagation, vol. 1, no. 1, pp. 37-64, Sept. 2025.

Literature review

Paper	Hardware implementation	Frequency / Waveform	Communication + Sensing arrangement	Limitation / Gap
Pegoraro *et al.*, "HiSAC: High-Resolution Sensing with Multiband Communication Signals"	RFSoC + 16-element Sivers phased arrays	58–64 GHz, multiband communication signals	Communication pilots are reused for sensing	Bands are all within the same mmWave region
Wang *et al.*, "An RFSoC-Based Scalable OTFS Prototyping Platform for Integrated Sensing and Communications"	RFSoC-based physical prototype	OTFS - Orthogonal Time Frequency Space	Same OTFS platform performs communication and sensing	Main flexibility is waveform/baseband
George *et al.*, "Hardware-Software Platform Enabling Joint Communication	Multi-purpose RX front-end, antennas and	5G NR n258, ~25 GHz, 1 GHz	Receiver front-end, antenna and waveform are	

Why multiple antenna sub arrays

- In antenna array the spacing between two antennas decided by the upper bound frequecy wave length of the desired band. (Sampling the wave phases without aliasing)
- Antenna array elements are equally spaced (linear array)

800MHz - 6000MHz band

d - space between two elements

$$d = \lambda/2$$

lambda - free space wave length of the upper bound frequency of the frequency band

$$\lambda_{6G} = \frac{c}{6 \times 10^9} = 50 \text{ mm}$$

$$\lambda_{800M} = \frac{c}{800 \times 10^6} = 375 \text{ mm}$$

- Therefore the spacial resolution for the lower frequencies is lower. For lower frequencies need a larger electrical aperture. Need a larger number of antenna elements.
- It is too costly

Solution :- Divide the the frequency range into proper sub ranges and allocate differnt antenna arrays.

https://en.wikipedia.org/wiki/Antenna_array

Estimation of link budget calculation for vivaldi antenna array

$$P_R = P_T G_T G_R \lambda^2 / (4\pi R)^2$$

PR Received power
PT Transmitted power
GT Transmitter antenna gain
R range
λ wavelength
GR Receive antenna gain

- Hopely recieve signals from 10-15 meters away

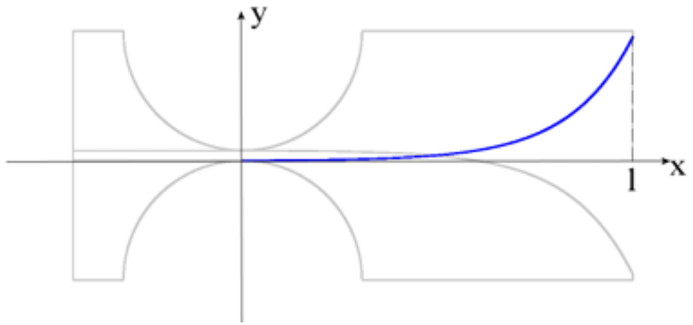
Free space path loss for 2.4 GHz frequency signal

FSPL = 20 log (4π(15)/0.125)
= **63.58 dB**

PR = PT + GT + GR −FSPL (In dBm)
PR = 6dB - 63.58dB

https://www.analog.com/en/resources/technical-articles/radio-linkbudget-calculations-for-ismrf-products.html?

Antipodal Vivaldi design



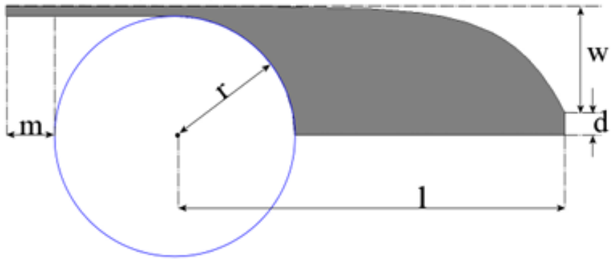


Fig. 2. Top metal layer in the Antipodal Vivaldi Antenna.

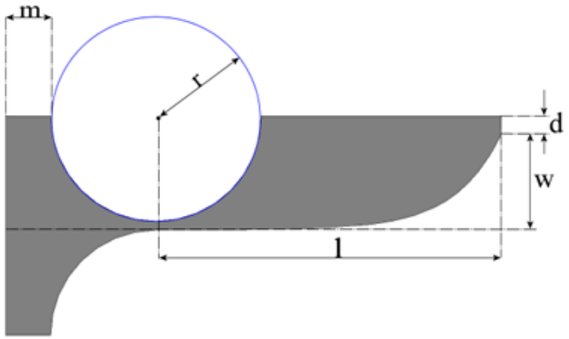
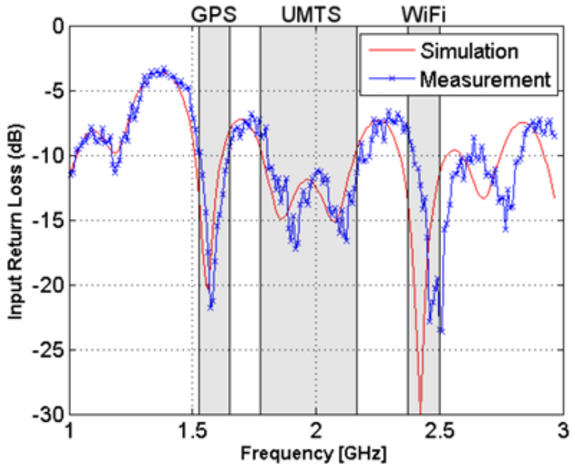


Fig. 3. Lower metal layer in the Antipodal Vivaldi Antenna.



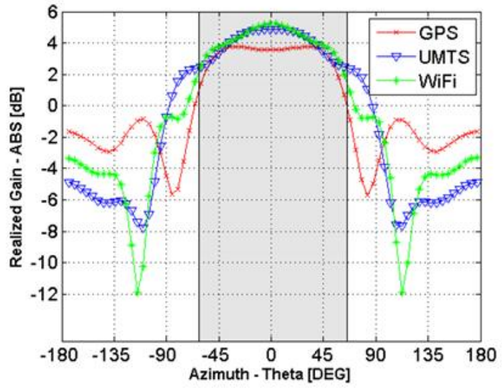


Fig. 7. Realized Gain vs. Azimuth.

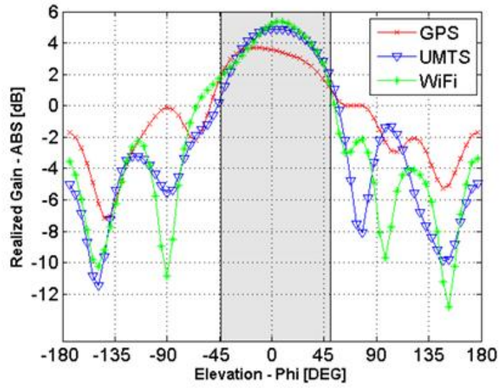
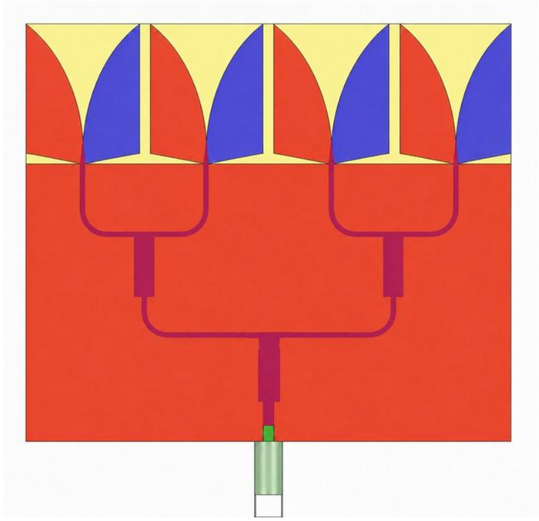


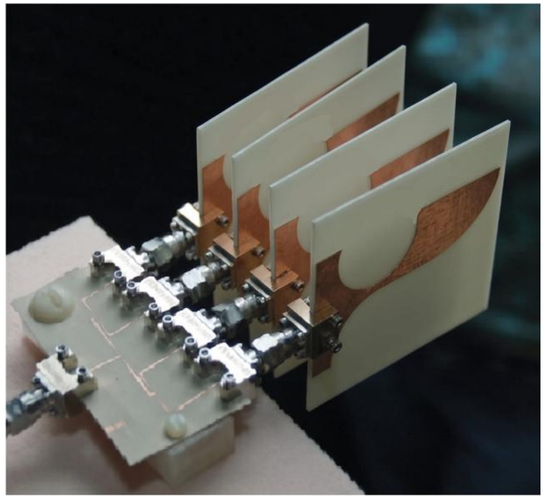
Fig. 8. Realized Gain vs. Elevation.

G. Vinci and R. Weigel, "Multiband Planar Vivaldi Antenna for Mobile Communication and Industrial Applications," 2010 International Conference on Electromagnetics in Advanced Applications, Sydney, NSW, Australia, 2010, pp. 93-96

Antipodal Vivaldi design



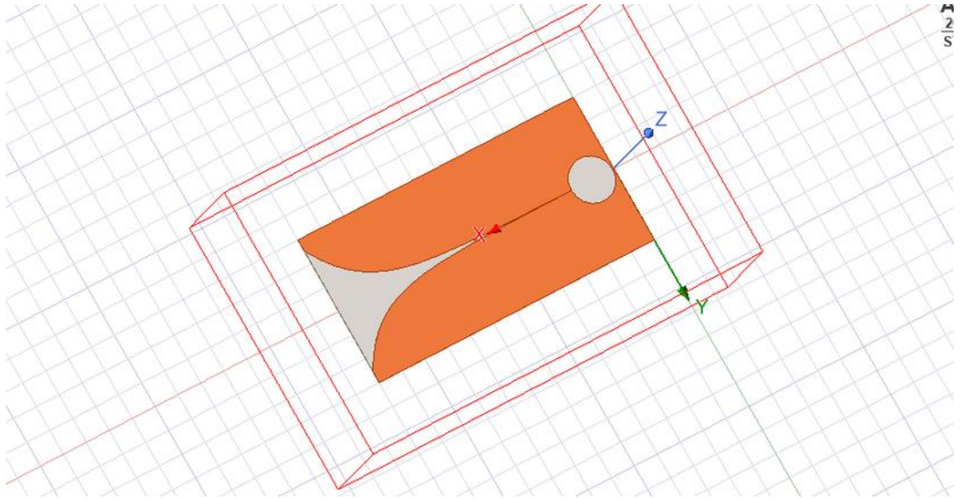
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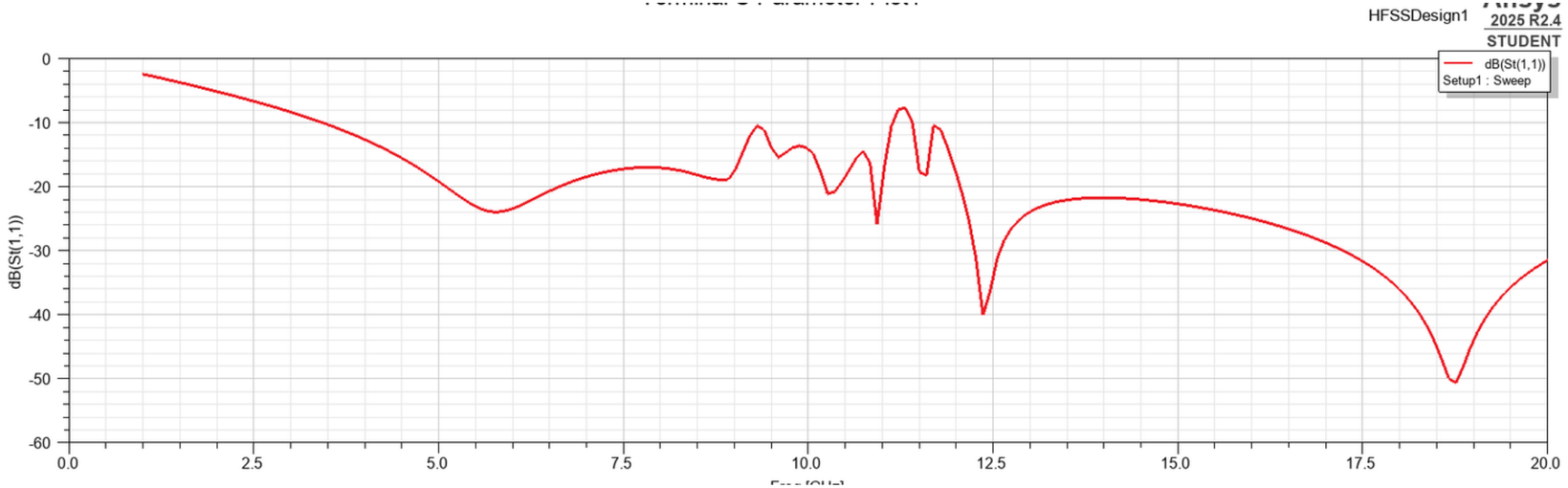
2

The second orientation is preferred because it allows the inter-element spacing d to be maintained close to the desired λ/2 spacing. In the first orientation, the physical dimensions of the Vivaldi elements constrain the placement, resulting in d>λ/2.

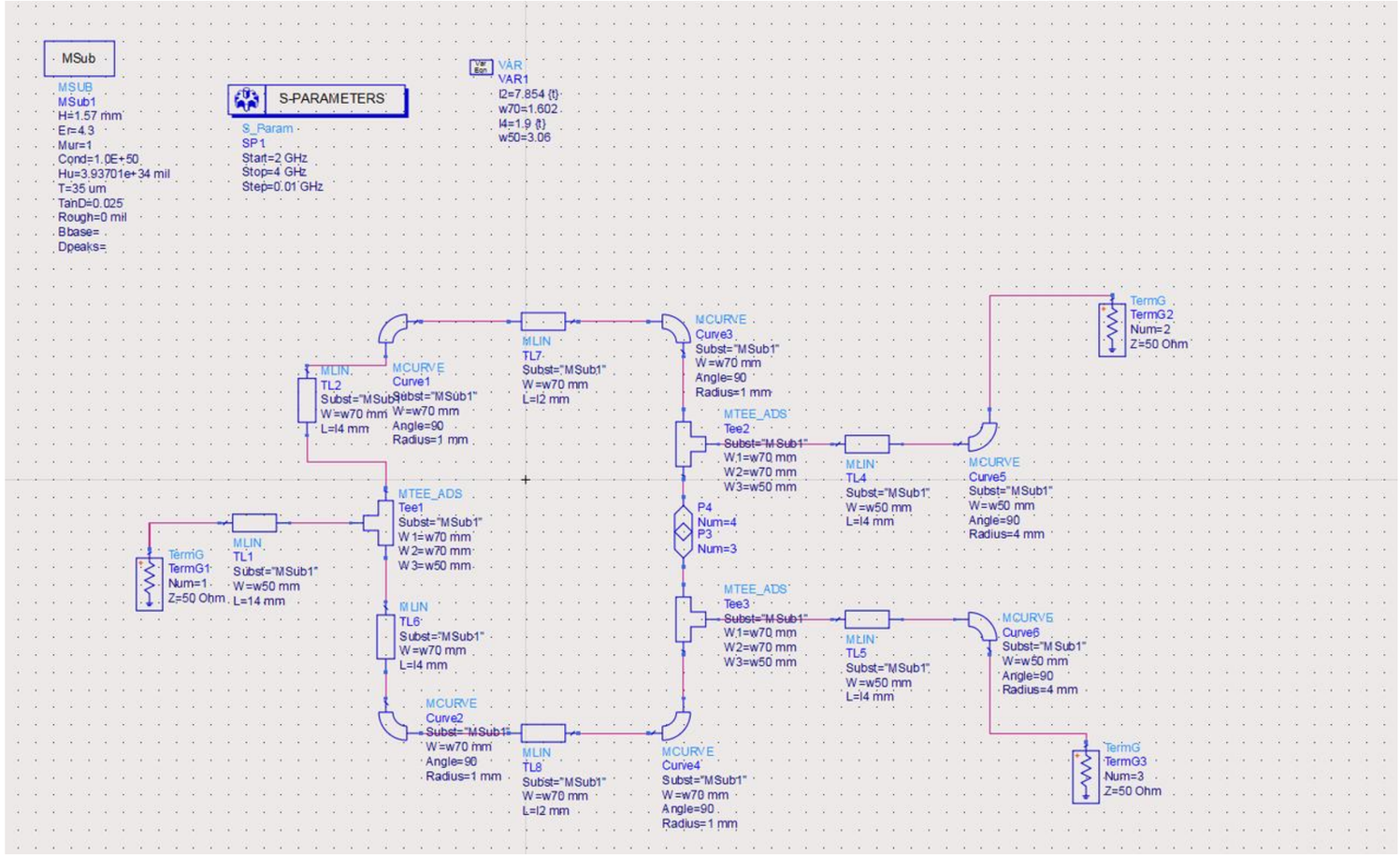
Tappered slot Vivaldi antenna simulation

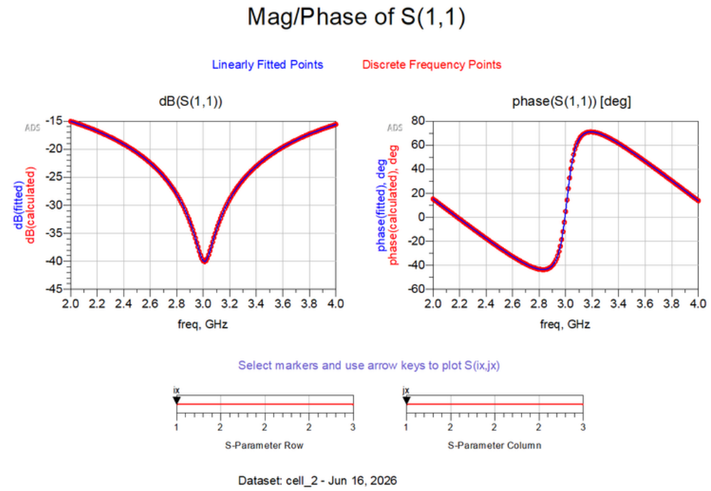


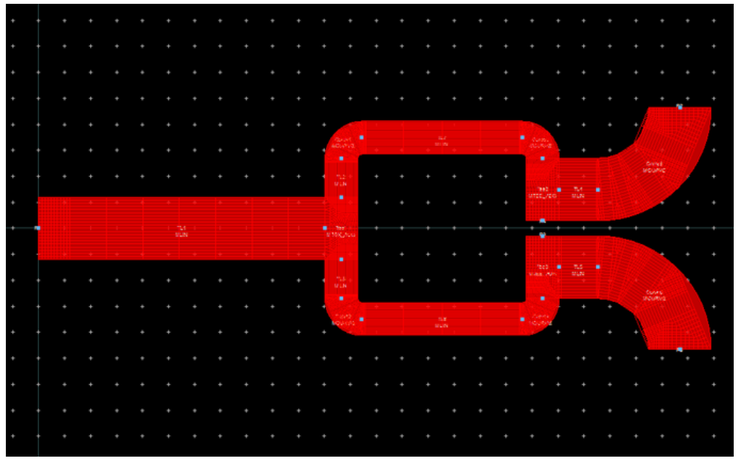
Name	Value	Unit	Evaluated V...	Type
s	0.015	mm	0.015mm	Design
rate	33.85		33.85	Design
h	0.508	mm	0.508mm	Design
TL	200	mm	200mm	Design
cThick	0.035	mm	0.035mm	Design
w	120		120	Design
a	120	mm	120mm	Design
L	240	mm	240mm	Design
MW	1.78	mm	1.78mm	Design



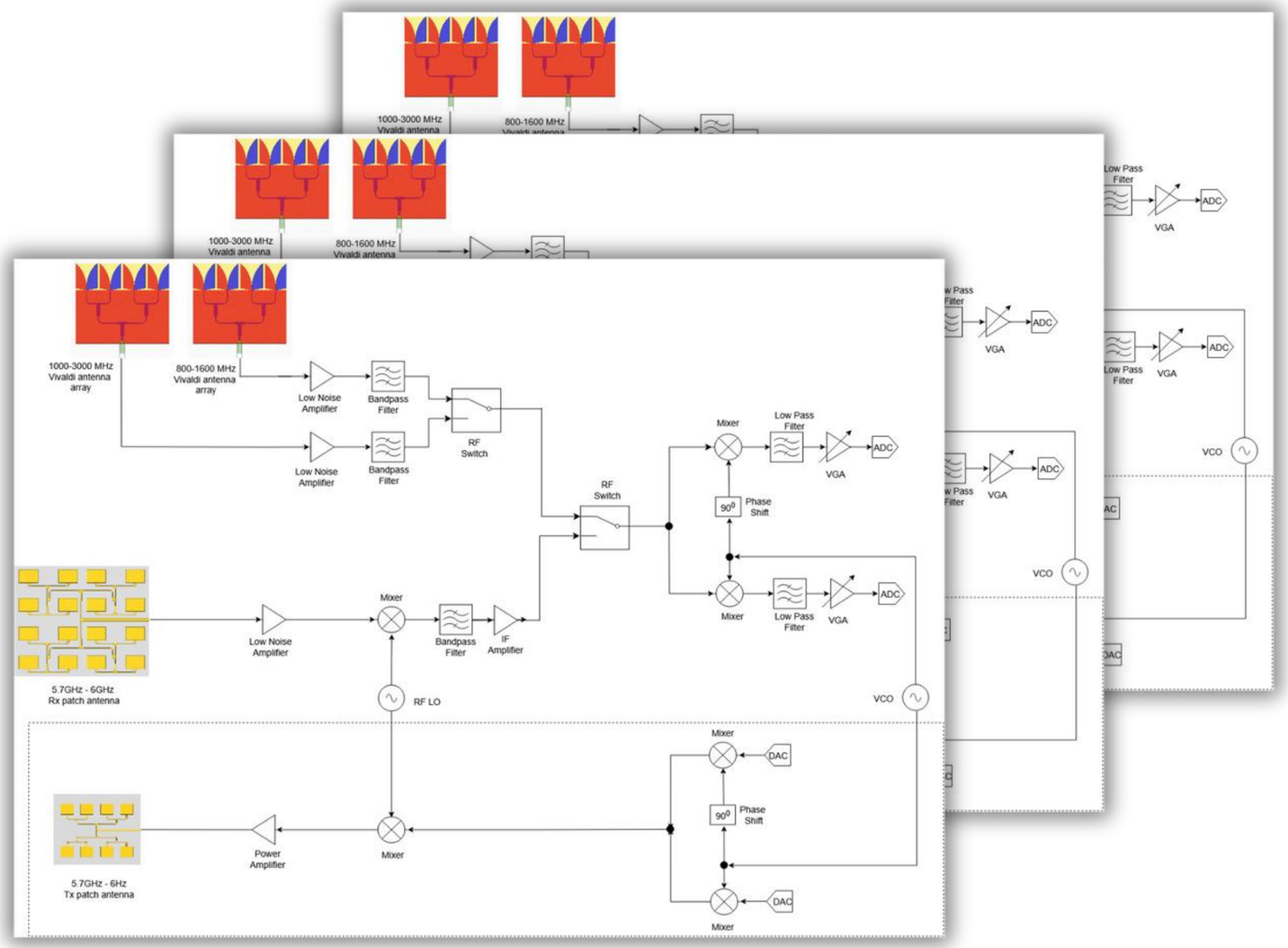
Wilkinson power divider simulation



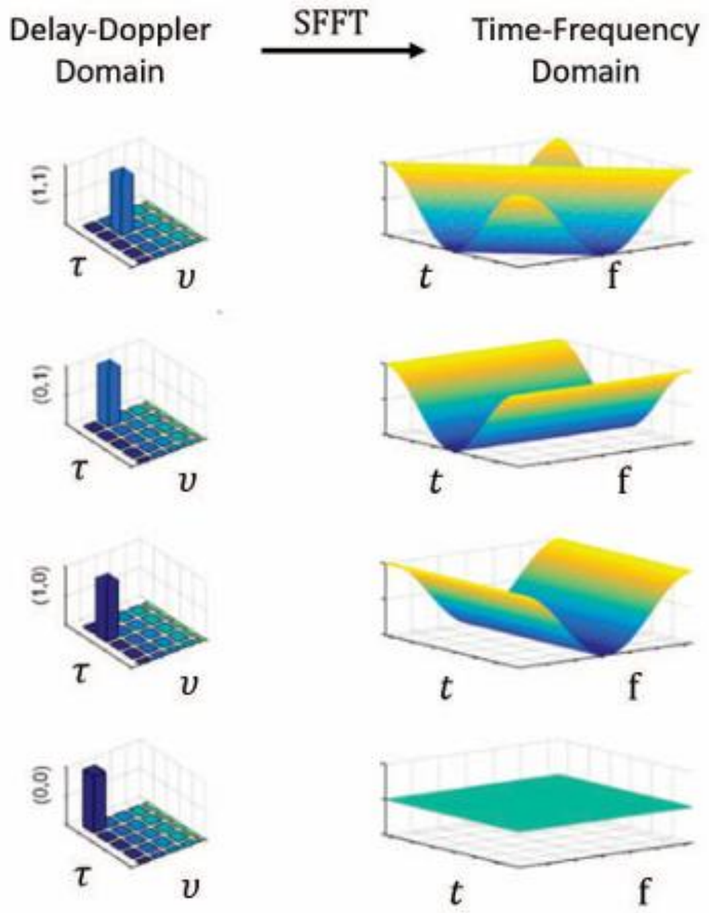




Reciever architecture for beamforming (RF front end)



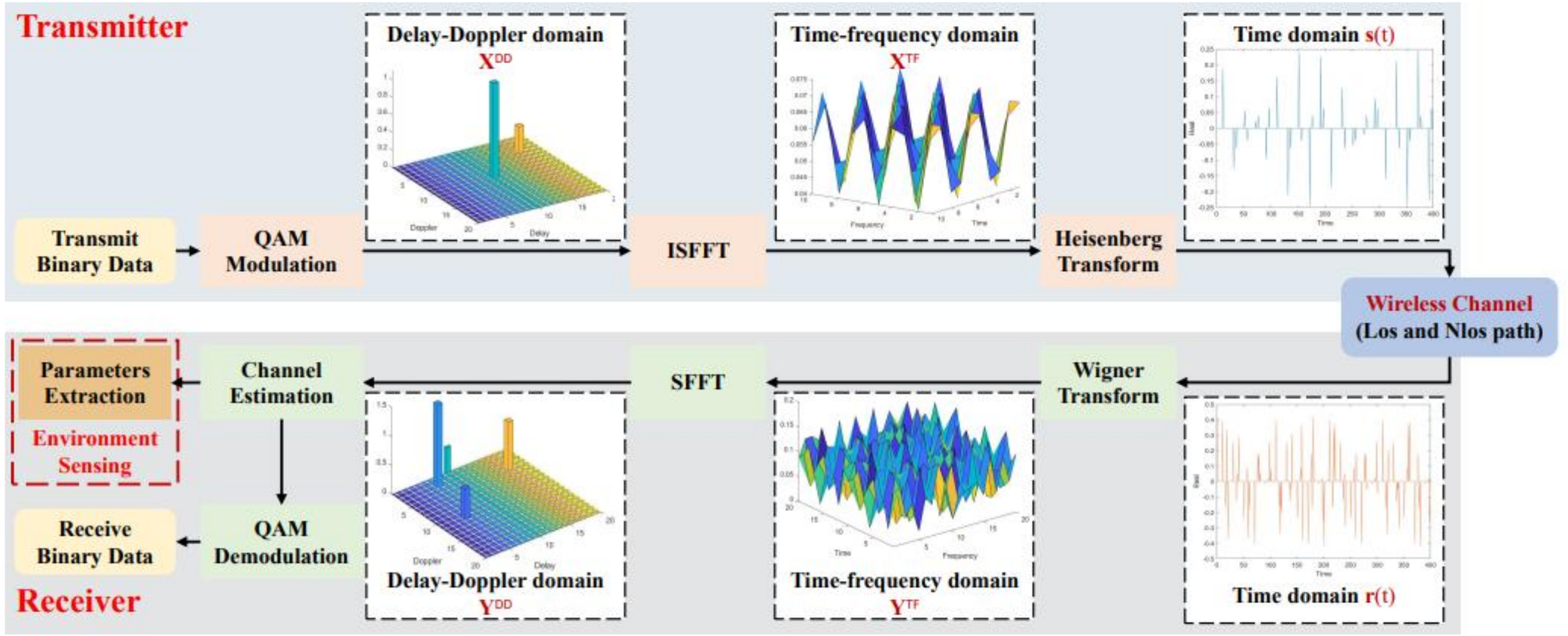
Delay Doppler domain and Time Frequency Domain



- This diagram shows how the **complex time frequency domain signal** is **simplified in the delay doppler domain**

R. Hadani et al., "Orthogonal Time Frequency Space Modulation," 2017 IEEE Wireless Communications and Networking Conference (WCNC), San Francisco, CA, USA, 2017, pp. 1-6.

OTFS Modulation Detailed Block diagram



J. Wu, Y. Shi, W. Yuan, Q. Cheng, B. Li and X. Wei, "SDR-Empowered Environment Sensing Design and Experimental Validation Using OTFS-ISAC Signals," in IEEE Transactions on Mobile Computing, vol. 24, no. 11, pp. 12252-12263, Nov. 2025.